

Last login: Wed Jun 17 18:49:21 on ttys000
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

```
tpch=# SET ROLE alice;
SET
tpch=> select
  o_orderpriority,
  count(*) as order_count
from
  orders
where
  o_orderdate >= date '1994-01-01'
  and o_orderdate < date '1994-01-01' + interval '3' month
  and exists (
    select
      *
    from
      lineitem
    where
      l_orderkey = o_orderkey
      and l_commitdate < l_receiptdate
  )
group by
  o_orderpriority
order by
  o_orderpriority;
WARNING:  Log 1: Planner Input:
DETAIL:  (QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname orders
  :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
  }
  :rtekind 0
  :relid 16561
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals (
    {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :opernName <>
    :subselect
    {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :cteList <>
    :rtable (
      {RANGETBLENTTRY
      :alias
      {ALIAS
      :aliasname c
      :colnames <>
      }
      :eref
      {ALIAS
      :aliasname c
      :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
      }
      :rtekind 0
      :relid 16546
      :inh true
      :relkind r
      :rellockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
      }
    )
  )
  :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 16546
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
    }
  )
  )
:jointree
{FROMEXPR
:fromList (
  {RANGETBLREF
  :rtindex 1
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 1
  }
  )
  }
  )
}
```

c.c_custkey = orders.o_custkey (First condition of RLS Policy)

```
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXPR
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -128 19 0 0 0 0 0 0 ]
}
)
:location -1
}
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 11
}
)
```

c.c_acctbal > 5000

```

)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 82
}
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 102
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 94
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 123
}
{OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 142
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1106
:conststypmod 19607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 166
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 155
}
)
:location 135
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStat <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu

```

Here we have user query quals

`o_orderdate >= DATE '1994-01-01'`

`o_orderdate < DATE '1994-01-01' + INTERVAL '3 months' =>
o_orderdate < DATE '1994-04-01'`

```

        mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
        "l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate"
        "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
        t")
    }
    :rtekind 0
    :relid 16566
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
})
}
:joinrees
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
})
}
:squals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 282
})
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
})
}
:location 293
})
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 322
})
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 337
})
}
:location 335
})
}
:location 318
})
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 220
})
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
})
{TARGETENTRY
:expr
{VAR
:varno 1

```

$l_orderkey = o_orderkey$

$l_commitdate < l_receiptdate$

```

:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 220
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 220
}
}
:resno 3
:resname 1_supplekey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 220
}
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 220
}
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 220
}
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 220
}
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 220
}
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8

```

```

:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 220
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 220
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 220
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 220
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 220
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 220
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 220
}

```

```

    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 16
  :vartype 25
  :vartypmod -1
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 16
  :location 220
  }
  :resno 16
  :resname l_comment
  :ressortgroupref 0
  :resorigtbl 16566
  :resorigcol 16
  :resjunk false
  }
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 184
}
:location 119
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList {
  {TARGETENTRY
  :expr
  {VAR
  :varno 2
  :varattno 1
  :vartype 25
  :vartypmod -1
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 2
  :varattnosyn 1
  :location -1
  }
  :resno 1
  :resname o_orderpriority
  :ressortgroupref 1
  :resorigtbl 16561
  :resorigcol 6
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {AGGREF
  :aggfnoid 2803
  :aggtype 28
  :aggcollid 0
  :inputcollid 0
  :aggtranstype 0
  :aggargtypes <>
  :aggdirectargs <>
  :args <>
  :aggorder <>
  :aggdistinct <>
  :aggfilter <>
  :aggstar true
  :aggvariadic false
  :aggkind n
  :aggpresorted false
  :agglevelsup 0
  :aggssplit 0
  :aggno -1
  :aggtransno -1
  :location 32
  }
  :resno 2
  :resname order_count
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause {
  {SORTGROUPCLAUSE
  :tleSortGroupRef 1
  :eqop 98
  :sortop 664
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
}
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause {
  {SORTGROUPCLAUSE
  :tleSortGroupRef 1
  :eqop 98
  :sortop 664
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
}
:limitOffset <>

```

GROUP BY o_orderpriority
ORDER BY o_orderpriority

```

:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 414
}

WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
}
:rtkind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>

```

```

:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 674 :opfncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtyp 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtyp 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})}
WARNING: max_parallel_hazard_walker: Param paramid=0 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 674 :opfncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtyp 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtyp 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe
0 = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL: (PATH
:pathtype 339
:parent_relds (b 1)
:pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs ()
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
})
:required_outer (b)
:parallel_aware false
:parallel_safe false
:parallel_workers 0
:rows 1
:disabled_nodes 0
:startup_cost 0
:total_cost 5922
:pathkeys <>
)

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: (QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:reldid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reldid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 674
:opfncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
(VAR
:varno 1

```

```

:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
)
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 674 :opfncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 674 :opfncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL:
{PATH
:pathtype 339
:parent_relids (b 1)
:pathtarget
{PATHTARGET
:exprs (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:sortgrouprefs ( 0)
:cost.startup 0
:cost.per_tuple 0
:width 4
:has_volatile_expr 0
}
:required_outer (b)
:parallel_aware false
:parallel_safe true
:parallel_workers 0
:rows 67808
:disabled_nodes 0
:startup_cost 0
:total_cost 5547
:pathkeys <>
}
)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
}

```

```

:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 0
    :relid 16561
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals ((
      {ALTERNATIVESUBPLAN
        :subplans (
          {SUBPLAN
            :subLinkType 0
            :testexpr <>
            :paramIds <>
            :plan_id 1
            :plan_name SubPlan\ 1
            :firstColType 2278
            :firstColTypmod -1
            :firstColCollation 0
            :useHashTable false
            :unknownEqFalse true
            :parallel_safe false
            :setParam <>
            :parParam (i 0)
            :args (
              {VAR
                :varno 1
                :varattno 2
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 2
                :location -1
              }
            )
            :startup_cost 0
            :per_call_cost 5922
          }
          {SUBPLAN
            :subLinkType 2
            :testexpr
              {OEXPR
                :opno 96
                :opfuncid 65
                :opresulttype 16
                :opreset false
                :opcollid 0
                :inputcollid 0
                :args (
                  {VAR
                    :varno 1
                    :varattno 2
                    :vartype 23
                    :vartypmod -1
                    :varcollid 0
                    :varnullingrels (b)
                    :varlevelsup 0
                    :varreturningtype 0
                    :varnosyn 1
                    :varattnosyn 2
                    :location -1
                  }
                )
                {PARAM
                  :paramkind 1
                  :paramid 1
                  :paramtype 23
                  :paramtypmod -1
                  :paramcollid 0
                  :location -1
                }
              }
            :location -1
          }
        )
        :paramIds (i 1)
        :plan_id 2
        :plan_name SubPlan\ 2
        :firstColType 23
        :firstColTypmod -1
        :firstColCollation 0
        :useHashTable true
        :unknownEqFalse true
        :parallel_safe true
        :setParam <>
        :parParam <>
        :args <>
        :startup_cost 5716.52
        :per_call_cost 0.0025
      }
    )
  )
)
)
{RANGETBLENTTRY
:alias <>
:eref
  {ALIAS
    :aliasname *GROUP*
    :colnames ("o_orderpriority")
  }
:rtekind 9
:groupexprs (
  {VAR
    :varno 1
    :varattno 6
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location 11
  }
)
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
  {ALIAS
    :aliasname lineitem

```

SubPlan 1
Correlated subplan
useHashTable false
Executed per outer row

SubPlan 2
Hashed subplan
c_custkey values hashed once
useHashTable true

securityQuals are not joined with the user query quals. One interpretation is that the planner has already converted the security quals into a SubPlan, so we no longer see the original predicates here.

```

:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_qualifiedprice" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:permindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 82
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 102
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 94
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 123
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 135
}
)
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}

```

orders SEMI JOIN lineitem
jointype 4 = SEMI JOIN

o_orderdate >= DATE '1994-01-01'

o_orderdate < TIMESTAMP '1994-04-01 00:00:00'

lineitem.l_orderkey = orders.o_orderkey

```

    }
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 295
    }
  )
  :location 293
  }
  {OPEXPR
  :opno 1095
  :opfuncid 1087
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
  {VAR
  :varno 3
  :varattno 12
  :vartype 1082
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 3
  :varattnosyn 12
  :location 322
  }
  {VAR
  :varno 3
  :varattno 13
  :vartype 1082
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 3
  :varattnosyn 13
  :location 337
  }
  )
  :location 335
  }
  )
  :alias <>
  :rtindex 0
  }
  )
  :quals <>
  }
:mergeActionList <>
:mergeFargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 32
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sorttop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sorttop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)

```

lineitem.l_commitdate < lineitem.l_receiptdate

```
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 414
}
```

In the first WARNING it is checking for the securityQuals whether it is safe to execute in parallel but at the end parallel_safe = 0 means flase because it has correlated_subplan

```
WARNING:  is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ((SUBPLAN :subLinkType 0 :testexpr <> :paramIds <> :plan_id 1 :plan_name SubPlan\ 1 :firstColType 2278 :fir
tColTypeMod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 0) :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :var
nullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) :startup_cost 0 :per_call_cost 5922) {SUBPLAN :subLinkType 2 :testexpr {OPEXPR :opno 96 :opfncid 65 :opresul
ttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn
n 2 :location -1) {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :paramIds (i 1) :plan_id 2 :plan_name SubPlan\ 2 :firstColType 23 :firstColTypeM
od -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 5716.52 :per_call_cost 0.0025})) :is_pushed_down true :can_join fa
lse :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :route
_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :oute
r_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0 {RESTRICTINFO :clause {OPEXPR :opno 1098 :
opfncid 1090 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 5 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0
:varnosyn 1 :varattnosyn 5 :location -1) {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]})
:location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :route_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_c
ost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvf
req -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING:  max_parallel_hazard_walker: SubPlan plan_id=1 parallel_safe=0 → reason=correlated_subplan
WARNING:  is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ((SUBPLAN :subLinkType 0 :testexpr <> :paramIds <> :plan_id 1 :plan_name SubPlan\ 1 :firstColType 2278 :fir
stColTypeMod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam <> :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :va
rnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) :startup_cost 0 :per_call_cost 5922) {SUBPLAN :subLinkType 2 :testexpr {OPEXPR :opno 96 :opfncid 65 :opresu
lttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattn
osyn 2 :location -1) {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :paramIds (i 1) :plan_id 2 :plan_name SubPlan\ 2 :firstColType 23 :firstColTypeM
od -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 5716.52 :per_call_cost 0.0025})) :is_pushed_down true :can_join f
alse :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :route
_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :out
er_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0 {RESTRICTINFO :clause {OPEXPR :opno 1098 :
opfncid 1090 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 5 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype
0 :varnosyn 1 :varattnosyn 5 :location -1) {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]})
:location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :requir
ed_relds (b 1) :incompatible_relds (b) :route_relds (b) :left_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em
<> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0 {RESTRICTINFO :clause {OPEXPR :opno 2345 :opfncid 2338 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 5 :vartype 1082 :vartypmod -1 :varcollid 0 :varn
ullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1) {CONST :consttype 1114 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location
-1 :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level
1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :route_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_c
ost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvf
req -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING:  is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1095 :opfncid 1087 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varattno 12 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 12 :location -1) {VAR :varno 3 :varattno 13 :vartype 1082 :vartypmod -1 :varcollid 0 :varnu
llingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 13 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :lea
kproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :route_relds (b) :left_relds (b 3) :right_relds (b 3) :orclause <>
:rinfo_serial 5 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :rig
ht_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING:  is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1095 :opfncid 1087 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varattno 12 :vartype
1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 12 :location -1) {VAR :varno 3 :varattno 13 :vartype 1082 :vartypmod -1 :varcollid 0 :varn
ullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 13 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :le
akproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :route_relds (b 3) :right_relds (b 3) :orclause <>
:rinfo_serial 5 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :rig
ht_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1)
)
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1
)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -
1) {AGGREF :aggfnoid 2083 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false
:aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) {AGGREF :aggfnoid 2083 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false
:aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -
1) {AGGREF :aggfnoid 2083 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false
:aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) {AGGREF :aggfnoid 2083 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false
:aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -
1)
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  Log 3: At the starting of Plan creation
DETAIL:  (AGGPATH
:path.pathtype 365
:parent_relds (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 5
:path.disabled_nodes 0
:path.startup_cost 8883274415.76885
:path.total_cost 8883274719.76382
:path.pathkeys {
:PATHKEY
:pk_eclass
{EQUIVALENCECLASS
:ec_opfamilies (o 1994 2095)
:ec_collation 100
:ec_childmembers_size 0
:ec_members {
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:em_relds (b 1)
:em_is_const false
:em_is_child false
:em_datatype 25
:em_jdomain
{JOINDOMAIN
:jd_relds (b 1 3)
}
}
}
:ec_childmembers <>
:ec_sources <>
:ec_derives_list <>
:ec_relds (b 1)
:ec_has_const false
:ec_has_volatile false
:ec_broken false
:ec_sortref 1
:ec_min_security 4294967295
:ec_max_security 0
}
}
:pk_opfamily 1994
:pk_cmptype 1
:pk_nulls_first false
}
)
:subpath
{SORTPATH
```

```

: path.pathtype 362
: parent_relids (b)
: path.pathtarget
{PATHTARGET
  :exprs (
    {VAR
      :varno 1
      :varattno 6
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 6
      :location -1
    }
  )
  :sortgrouprefs (1)
  :cost.startup 0
  :cost.per_tuple 0
  :width 9
  :has_volatile_expr 0
}
: required_outer (b)
: path.parallel_aware false
: path.parallel_safe false
: path.parallel_workers 0
: path.rows 40534
: path.disabled_nodes 0
: path.startup_cost 8883274415.708885
: path.total_cost 8883274517.043884
: path.pathkeys (
  {PATHKEY
    :pk_eclass
    {EQUIVALENCECLASS
      :ec_opfamilies (o 1994 2095)
      :ec_collation 100
      :ec_childmembers_size 0
      :ec_members (
        {EQUIVALENCEMEMBER
          :em_expr
          {VAR
            :varno 1
            :varattno 6
            :vartype 25
            :vartypmod -1
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 6
            :location -1
          }
          :em_relids (b 1)
          :em_is_const false
          :em_is_child false
          :em_datatype 25
          :em_jdomain
          {JOINDOMAIN
            :jd_relids (b 1 3)
          }
        }
      )
      :ec_childmembers <>
      :ec_sources <>
      :ec_derives_list <>
      :ec_relids (b 1)
      :ec_has_const false
      :ec_has_volatile false
      :ec_broken false
      :ec_sortref 1
      :ec_min_security 4294967295
      :ec_max_security 0
    }
    :pk_opfamily 1994
    :pk_cmptype 1
    :pk_nulls_first false
  }
)
: subpath
{HASHPATH
  :jpath.pathtype 359
  :parent_relids (b 1 3)
  :jpath.pathtarget
  {PATHTARGET
    :exprs (
      {VAR
        :varno 1
        :varattno 6
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
      }
    )
    :sortgrouprefs (1)
    :cost.startup 0
    :cost.per_tuple 0
    :width 9
    :has_volatile_expr 0
  }
  : required_outer (b)
  :jpath.path.parallel_aware false
  :jpath.path.parallel_safe false
  :jpath.path.parallel_workers 0
  :jpath.path.rows 40534
  :jpath.path.disabled_nodes 0
  :jpath.path.startup_cost 212549.3375
  :jpath.path.total_cost 8883271313.470623
  :jpath.path.pathkeys <>
  :jpath.jointype 4
  :jpath.inner_unique false
  :jpath.outerjoinpath
  {PATH
    :pathtype 339
    :parent_relids (b 1)
    :required_outer (b)
    :parallel_aware false
    :parallel_safe false
    :parallel_workers 0
    :rows 144199
    :disabled_nodes 0
    :startup_cost 0
    :total_cost 8883048508
    :pathkeys <>
  }
  :jpath.innerjoinpath
  {PATH
    :pathtype 339
    :parent_relids (b 3)
    :required_outer (b)
    :parallel_aware false
    :parallel_safe true
    :parallel_workers 0
    :rows 2000307
    :disabled_nodes 0
    :startup_cost 0
    :total_cost 179731.5
    :pathkeys <>
  }
}

```

```

:joinrestrictinfo (
{RESTRICTINFO
:clause
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
)
:location -1
}
:is_pushed_down true
:can_join true
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 2
:security_level 1
:num_base_rels 2
:clause_relds (b 1 3)
:required_relds (b 1 3)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b 1)
:right_relds (b 3)
:orclause <>
:rinfo_serial 6
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 6.666666666666667e-07
:outer_selec 0.28109466666666666
:mergeopfamilies (o 1976)
:left_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
:em_relds (b 1)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relds (b 1 3)
}
}
:right_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
:em_relds (b 3)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relds (b 1 3)
}
}
:outer_is_left false
:hashjoinoperator 96
:left_bucketsize 6.934860852017004e-06
:right_bucketsize 7.115057596391243e-06
:left_mcvfreq 0
:right_mcvfreq 0
:left_hasheqoperator 96
:right_hasheqoperator 96
}
)
:path_hashclauses (
{RESTRICTINFO
:clause
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
{VAR
:varno 3

```

```

:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
}
:location -1
}
:is_pushed_down true
:can_join true
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 2
:security_level 1
:num_base_rels 2
:clause_relds (b 1 3)
:required_relds (b 1 3)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b 1)
:right_relds (b 3)
:orclause <
:rinfo_serial 6
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 6.666666666666667e-07
:outer_selec 0.28109466666666666
:mergeopfamilies (o 1976)
:left_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
:em_relds (b 1)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relds (b 1 3)
}
}
:right_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
:em_relds (b 3)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relds (b 1 3)
}
}
:outer_is_left false
:hashjoinoperator 96
:left_bucketsize 6.934860852017004e-06
:right_bucketsize 7.115057596391243e-06
:left_mcvfreq 0
:right_mcvfreq 0
:left_hasheqoperator 96
:right_hasheqoperator 96
}
}
:num_batches 16
:inner_rows_total 2000307
}
}
:aggstrategy 1
:aggssplit 0
:numGroups 5
:transitionSpace 0
:groupClause {
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sorstop 664
:reverse_sort false
:nulls_first false
:hashable true
}
}
}
:qual <>
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 31
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 8883274415.708885
:plan.total_cost 8883274719.763882
:plan.plan_rows 5
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist {
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1

```

```

:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtransype 20
:aggartypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno 0
:aggtransno 0
:location 32
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:plan.qual <>
:plan.lefttree
{SORT
:plan.disabled_nodes 0
:plan.startup_cost 8883274415.708885
:plan.total_cost 8883274517.043884
:plan.plan_rows 40534
:plan.plan_width 9
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 1
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 212549.3375
:join.plan.total_cost 8883271313.470623
:join.plan.plan_rows 40534
:join.plan.plan_width 9
:join.plan.parallel_aware false
:join.plan.parallel_safe false
:join.plan.async_capable false
:join.plan.plan_node_id 2
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:join.plan.qual <>
:join.plan.lefttree
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 8883048508
:scan.plan.plan_rows 144199
:scan.plan.plan_width 13
:scan.plan.parallel_aware false
:scan.plan.parallel_safe false
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1

```

```

:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:scan.plan.qual (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 82
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 102
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 94
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramIds (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColltype 23
:firstColltypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 5716.52
:per_call_cost 0.0025
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 123
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 135
}

```

```

)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 1
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 179731.5
:plan.total_cost 179731.5
:plan.plan_rows 2000307
:plan.plan_width 4
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 4
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 179731.5
:scan.plan.plan_rows 2000307
:scan.plan.plan_width 4
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 5
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 322
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 337
}
)
:location 335
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
)
:skewTable 16561
:skewColumn 1
:skewInherit false
:rows_total 0

```

```

    }
:join.plan.initPlan <>
:join.plan.extParam (b)
:join.plan.allParam (b)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
  {OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno -2
:varattno 2
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
  {VAR
:varno -1
:varattno 1
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
)
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
  {VAR
:varno -2
:varattno 2
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:aggstrategy 1
:aggssplit 0
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rttable (
  {RANGETBLENTY
:alias <>
:eref
  {ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
  {RANGETBLENTY
:alias <>
:eref
  {ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs <>
:lateral false
:inFromCl false
:securityQuals <>
}
  {RANGETBLENTY
:alias <>
:eref
  {ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
  {RANGETBLENTY
:alias
  {ALIAS

```

```

        :aliasname c
        :colnames <>
    }
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:unprunableRelids (b 1 3 4 5)
:permInfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
}
:resultRelations <>
:appendRelations <>
:subplans (<>
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 5547
:scan.plan.plan_rows 67808
:scan.plan.plan_width 4
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 6
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:scan.plan.qual (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true

```

```

        :constisnull false
        :location -1
        :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
    }
    :location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanreliid 5
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 16561 16566 16546 16546)
:invalidItems <>
:paramExecTypes (o 23 23)
:utilityStmt <>
:stmt_location 0
:stmt_len 414
}

```

o_orderpriority order_count	
1-URGENT	4751
2-HIGH	4737
3-MEDIUM	4661
4-NOT SPECIFIED	4591
5-LOW	4577
(5 rows)	

tpch=>